

3

Linear Operators on Vector Spaces

In the previous chapter, we discussed bases and the methods for changing the basis representation of a vector. This change of basis was performed by multiplying the representation (the n -tuple of components) with an appropriate matrix, which described the relationship between the two sets of basis vectors [see Equation (2.33)]. When considering the change of basis operation, it is important to remember that the vector itself does not change, only the coordinate system in which it is expressed.

In this chapter, we will consider linear operators. Linear operators are functions on the vector space but are fundamentally different from the change of basis, although they will also be expressed in terms of a matrix multiplication. A linear operator, or linear transformation, is a process by which a given vector is transformed into an entirely different vector. As we shall see, linear operators might also implicitly perform a change of basis, they might take a vector in one space and produce a vector in an entirely different space, or they may simply change a vector in a given space into a different vector in the same space.

3.1 Definition of a Linear Operator

A linear operator is simply a rule that associates one vector with a different vector, which may or not be in the same linear vector space. It also must satisfy the properties of linearity. To be exact, we give the following definition:

Linear Operator (Transformation): An operator A from linear vector space $\mathbf{X}$ to linear vector space $\mathbf{Y}$, denoted $A: \mathbf{X} \rightarrow \mathbf{Y}$, is linear if

$$A(\alpha_1 \mathbf{x}_1 + \alpha_2 \mathbf{x}_2) = \alpha_1 A\mathbf{x}_1 + \alpha_2 A\mathbf{x}_2 \quad (3.1)$$

for any $\mathbf{x}_1, \mathbf{x}_2 \in \mathbf{X}$, and scalars α_1 and α_2 .

3.1.1 Range and Null Spaces

A linear operator such as the one in the definition above, $A: \mathbf{X} \rightarrow \mathbf{Y}$, will produce a vector $\mathbf{y} \in \mathbf{Y}$ and can be written as $A\mathbf{x} = \mathbf{y}$. The vector $\mathbf{y}$ produced by this operation is said to be in the *range* of operator A and is sometimes referred to as the *image* of $\mathbf{x}$ in $\mathbf{Y}$ ($\mathbf{x}$ being the *preimage* of $\mathbf{y}$). The concept of the range of an operator is similar to that of scalar functions, but linear operators in general will have multidimensional ranges. These ranges will themselves be linear vector spaces.

Range Space: The range space of an operator $A: \mathbf{X} \rightarrow \mathbf{Y}$, denoted $R(A)$, is the set of all vectors $\mathbf{y}_i \in \mathbf{Y}$ such that for every $\mathbf{y}_i \in R(A)$ there exists an $\mathbf{x} \in \mathbf{X}$ such that $A\mathbf{x} = \mathbf{y}_i$.
(3.2)

If the range of operator A is the entire space $\mathbf{Y}$, then A is said to be *onto*, or *surjective*. If A maps elements of $\mathbf{X}$ to *unique* values in $\mathbf{Y}$, i.e., if $\mathbf{x}_1 \neq \mathbf{x}_2$ implies that $A(\mathbf{x}_1) \neq A(\mathbf{x}_2)$, then A is said to be *one-to-one*, or *injective*. If operator A is both one-to-one and onto, then it is *bijective*, in which case it is also *invertible*.^M An operator A is invertible if and only if there exists another operator, denoted $A^{-1}: \mathbf{Y} \rightarrow \mathbf{X}$ such that $A^{-1}(A(\mathbf{x})) = \mathbf{x}$ and $A(A^{-1}(\mathbf{y})) = \mathbf{y}$. Then $A^{-1}A$ is denoted $I_{\mathbf{X}}$ and AA^{-1} is denoted $I_{\mathbf{Y}}$, which are the *identity*^M operators in their respective spaces. When A maps a space into itself, as in $A: \mathbf{X} \rightarrow \mathbf{X}$, we write simply $AA^{-1} = A^{-1}A = I$.

We will often need to know which, among all vectors $\mathbf{x} \in \mathbf{X}$, will map to the zero vector in $\mathbf{Y}$. For this reason we define the null space^M of an operator.

$\text{inv}(A)$

$\text{eye}(n)$

$\text{null}(A)$

Null Space: The null space of operator A , denoted $N(A)$, is the set of all vectors $\mathbf{x}_i \in \mathbf{X}$ such that $A(\mathbf{x}_i) = \mathbf{0}$:

$$N(A) = \{\mathbf{x}_i \in \mathbf{X} | A\mathbf{x}_i = \mathbf{0}\} \quad (3.3)$$

Example 3.1: The Range and Null Space of a Projection

Consider the projection theorem given in Chapter 2. Show that such a projection $P: \mathfrak{R}^3 \rightarrow U$ is a linear operator and find its range and null spaces.

Solution:

Suppose we have two vectors $\mathbf{x}_1, \mathbf{x}_2 \in \mathbf{X}$ and two real scalars α_1 and α_2 . Then let $\mathbf{x} = \alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2 \in \mathbf{X}$. The projection theorem says that there exists a vector $\mathbf{u} \in U$ such that if we denote the projection $\mathbf{u} = P\mathbf{x}$, then $\langle \mathbf{x} - P\mathbf{x}, \mathbf{y} \rangle = 0$ for all $\mathbf{y} \in U$. So by the definition of linearity, we have

$$\begin{aligned} \langle \alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2 - P(\alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2), \mathbf{y} \rangle &= \langle \alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2, \mathbf{y} \rangle - \langle P(\alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2), \mathbf{y} \rangle \\ &= \alpha_1\langle \mathbf{x}_1, \mathbf{y} \rangle + \alpha_2\langle \mathbf{x}_2, \mathbf{y} \rangle - \langle P(\alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2), \mathbf{y} \rangle \\ &= 0 \end{aligned}$$

so

$$\langle P(\alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2), \mathbf{y} \rangle = \alpha_1\langle \mathbf{x}_1, \mathbf{y} \rangle + \alpha_2\langle \mathbf{x}_2, \mathbf{y} \rangle \quad (3.4)$$

But considering the projections $P\mathbf{x}_1$ and $P\mathbf{x}_2$ separately, we have $\langle \mathbf{x}_1 - P\mathbf{x}_1, \mathbf{y} \rangle = 0$ and $\langle \mathbf{x}_2 - P\mathbf{x}_2, \mathbf{y} \rangle = 0$, which give $\langle \mathbf{x}_1, \mathbf{y} \rangle = \langle P\mathbf{x}_1, \mathbf{y} \rangle$ and $\langle \mathbf{x}_2, \mathbf{y} \rangle = \langle P\mathbf{x}_2, \mathbf{y} \rangle$, or, equivalently by multiplying each term by a scalar, $\alpha_1\langle \mathbf{x}_1, \mathbf{y} \rangle = \alpha_1\langle P\mathbf{x}_1, \mathbf{y} \rangle$ and $\alpha_2\langle \mathbf{x}_2, \mathbf{y} \rangle = \alpha_2\langle P\mathbf{x}_2, \mathbf{y} \rangle$. We can substitute these expressions into (3.4) above, to give

$$\begin{aligned} \langle P(\alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2), \mathbf{y} \rangle &= \alpha_1\langle P\mathbf{x}_1, \mathbf{y} \rangle + \alpha_2\langle P\mathbf{x}_2, \mathbf{y} \rangle \\ &= \langle \alpha_1P\mathbf{x}_1 + \alpha_2P\mathbf{x}_2, \mathbf{y} \rangle \end{aligned} \quad (3.5)$$

Recalling that this must be true for all $\mathbf{y} \in U$, then comparing (3.4) and (3.5) gives $P(\alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2) = \alpha_1P\mathbf{x}_1 + \alpha_2P\mathbf{x}_2$ and linearity is proven. The relationships between all these vectors are shown in Figure 3.1.

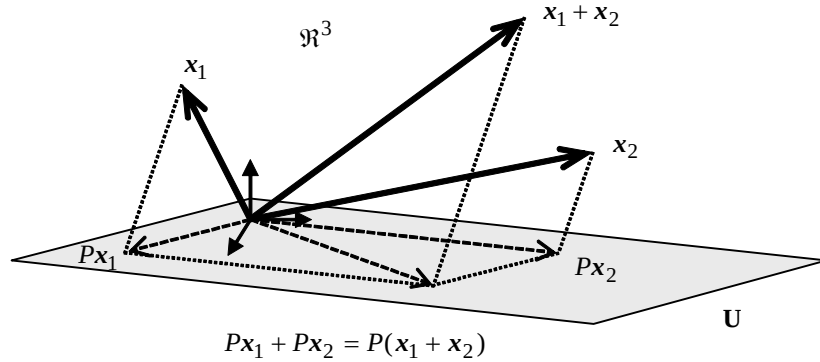


Figure 3.1 Graphical representation of the linearity of the projection operator. The projection of the sum of vectors is the sum of the projections of the vectors.

The range space of operator P , $R(P)$, is obviously the entire subspace $\mathbf{U}$. This is apparent from the observation that every vector in the entire subspace will have a preimage that can be constructed by adding to it any vector orthogonal to the plane $\mathbf{U}$. One can see that, in fact, every vector in $\mathbf{U}$ will have infinitely many preimages because there are infinitely many such orthogonal vectors that can be used. The range space therefore has dimension two, the same as the dimension of $\mathbf{U}$ itself.

The null space $N(P)$ consists of the set of vectors in $\mathbf{X}$ that project onto the zero vector of $\mathbf{U}$. These must therefore be constructed by taking all vectors orthogonal to $\mathbf{U}$ and adding them to this zero vector, which is the origin of the plane. Suppose, for example, the subspace $\mathbf{U}$ is described by the equation of a plane $f(\mathbf{x}) = 5x_1 + 3x_2 - 2x_3 = 0$. Then the set of all vectors orthogonal to the plane consists of the set of vectors $\alpha \nabla f = \alpha [5 \ 3 \ -2]^T$, where α is any scalar and $\nabla(\cdot)$ is the gradient operator described in Appendix A. Therefore, $N(P)$ is the straight line passing through the origin, extending in the direction of the vector $[5 \ 3 \ -2]^T$. One may wish to verify this using the projection theorem and any vector (other than the zero vector) that lies in $\mathbf{U}$. Being a single line, the null space is therefore of dimension one.

We will next show that any finite-dimensional linear operator can be represented for computational purposes as a matrix, also called A , acting on the representations of vectors. For linear operator A , the range space $R(A)$ is a subspace of $\mathbf{Y}$ whose dimension is the rank of matrix A . Conversely, the null space $N(A)$ is a subspace of $\mathbf{X}$, whose dimension is the nullity of matrix A . It can be shown that not only can any operator be represented by a matrix, but any matrix can be interpreted as a linear operator. This correspondence will prevent any

confusion regarding the notation. We can use the same symbol A for both a matrix and an operator without ambiguity because they are essentially one and the same.

3.1.2 Matrix Representations of Linear Operators

For generality, we will discuss the matrix representation of linear operators that might also change the basis of the vector on which it operates. This is obviously necessary if we speak of an operator that operates on vectors in a space of one dimension and returns vectors in a different dimension. Clearly a single basis would not suffice to describe both vectors.

Consider two vectors, one from an n -dimensional space and one from an m -dimensional space: $\mathbf{x} \in \mathbf{X}^n$ and $\mathbf{y} \in \mathbf{X}^m$. We will derive a matrix representation A for an operator that transforms a vector from one space into a vector in the other, i.e., $A: \mathbf{X}^n \rightarrow \mathbf{X}^m$. Denote arbitrary bases for the two spaces as

$$\{\mathbf{v}_j\} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} \text{ for space } \mathbf{X}^n$$

and

$$\{\mathbf{u}_i\} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\} \text{ for space } \mathbf{X}^m$$

By expanding $\mathbf{x}$ out as a representation in its basis as $\mathbf{x} = \sum_{j=1}^n \alpha_j \mathbf{v}_j$ and applying the linearity property of the operator A ,

$$\mathbf{y} = A\mathbf{x} = A\left(\sum_{j=1}^n \alpha_j \mathbf{v}_j\right) = \sum_{j=1}^n \alpha_j A(\mathbf{v}_j) \quad (3.6)$$

This simple but important result means that we can determine the effect of A on any vector $\mathbf{x}$ by knowing only the effect that A has on the basis vectors in which $\mathbf{x}$ is written. Equation (3.6) can be alternatively written in vector-matrix notation as:

$$\mathbf{y} = A\mathbf{x} = \begin{bmatrix} A\mathbf{v}_1 & A\mathbf{v}_2 & \cdots & A\mathbf{v}_n \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}$$

Now we note that each vector $A\mathbf{v}_j$ is by definition a vector in the range space $\mathbf{X}^m$. Then like any other vector in that space, it can be expanded in the basis defined for that space, $\{\mathbf{u}_i\}$. This expansion gives

$$A\mathbf{v}_j = \sum_{i=1}^m a_{ij}\mathbf{u}_i \quad (3.7)$$

for some set of coefficients a_{ij} . We can also expand $\mathbf{y}$ out directly in terms of its own basis vectors, using coefficients β_i ,

$$\mathbf{y} = \sum_{i=1}^m \beta_i \mathbf{u}_i \quad (3.8)$$

Substituting (3.7) and (3.8) into Equation (3.6), we can write:

$$\mathbf{y} = \sum_{j=1}^n \alpha_j \left(\sum_{i=1}^m a_{ij} \mathbf{u}_i \right) = \sum_{i=1}^m \beta_i \mathbf{u}_i \quad (3.9)$$

By changing the order of summation in this expression, we obtain

$$\sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} \alpha_j \right) \mathbf{u}_i = \sum_{i=1}^m \beta_i \mathbf{u}_i \quad (3.10)$$

But because of the uniqueness of any vector's expansion in a basis [see Section (2.2.3)], the expansion of $\mathbf{y}$ in $\{\mathbf{u}_i\}$ must be unique. This implies that

$$\beta_i = \sum_{j=1}^n a_{ij} \alpha_j \quad \text{for all } i = 1, \dots, m \quad (3.11)$$

The reader may notice that the expression (3.11) above is in the form of a matrix-vector multiplication. In fact, this is how we will usually apply this result. If $\alpha = [\alpha_1 \ \alpha_2 \ \dots \ \alpha_n]^T$ is the representation of $\mathbf{x} \in \mathbf{X}^n$ in the $\{\mathbf{v}_j\}$ basis and $\beta = [\beta_1 \ \beta_2 \ \dots \ \beta_m]^T$ is the representation of $\mathbf{y} \in \mathbf{X}^m$ in the $\{\mathbf{u}_i\}$ basis, then we can find this representation of $\mathbf{y}$ with the matrix multiplication $\beta = A\alpha$, where we use our new matrix representation for operator A :

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \quad (3.12)$$

That is, the $(i, j)^{th}$ element of the $(m \times n)$ matrix A is the coefficient from (3.7), which describes how the operator A transforms the basis vectors from one space into the basis of the range space. Of course, this implies that the bases of each space must be specified before determining the matrix representation for any linear operator. In general, the following observation can be made from a comparison of (3.7) and (3.12):

The j^{th} column of the matrix representation of any linear operator $A: \mathbf{X}^n \rightarrow \mathbf{X}^m$ can be constructed as the representation of the vector that is the result of A acting on the j^{th} basis vector of $\mathbf{X}^n$, where this new vector is expanded into the basis of the range space. That is, the j^{th} column of A is simply $A\mathbf{v}_j$, written as a representation in the basis $\{\mathbf{u}_i\}$.

This is a very useful property, as we shall see in the subsequent examples. It allows us a convenient way to determine the matrix representation of an arbitrary linear operator. Use of the matrix representation will be necessary for any computational process, such as the ones we will use in future chapters.

The representation of the linear operator as a matrix also has implications for the computation of the range and null spaces. As we will demonstrate in Section 3.3, the linear operation $\mathbf{y} = A\mathbf{x}$ can be viewed numerically as

$$\begin{aligned} \mathbf{y} &= A\mathbf{x} \\ &= \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix} \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_n \end{bmatrix} \end{aligned} \quad (3.13)$$

where a_i denotes the i^{th} column of the matrix A and $\mathbf{x}_i$ is the i^{th} component of the vector $\mathbf{x}$ or of its representation. Therefore, if the range space is the space of all possible values of $A\mathbf{x}$, it may be represented as the span of all of the columns of A . This implies that the rank of matrix A is equal to the dimension of the range space of operator A , i.e., $r(A) = \dim(R(A))$. Similarly, the null space of operator A may be represented as the space of all solutions of the simultaneous linear equations

$$\begin{aligned} \mathbf{0} &= A\mathbf{x} \\ &= \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \end{aligned} \quad (3.14)$$

Therefore the dimension of the null space of *operator* A is equal to the nullity of *matrix* A , i.e., $q(A) = \dim(N(A))$. For Example 3.1, $r(P) = 2$ and $q(P) = 1$. See Section 3.3 for a complete discussion the computation of these spaces.

Example 3.2: Rotation Matrices

Consider a linear operator $A: \mathfrak{R}^2 \rightarrow \mathfrak{R}^2$ that takes a vector $\mathbf{x}$ and rotates it counterclockwise by an angle θ as shown in Figure 3.2. Find the matrix representation of the linear operator that performs such planar rotations on arbitrary vectors in $\mathfrak{R}^2$, and test this matrix by rotating the vector $\mathbf{x} = [1 \ 2]^T$ by an angle $\theta = 30^\circ$.

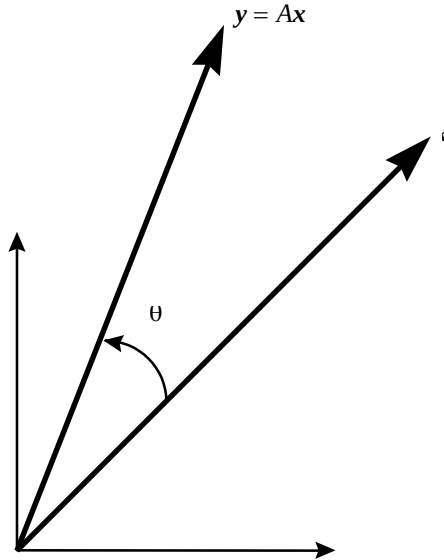


Figure 3.2

Vector $\mathbf{x}$ is transformed by operator A into a vector $\mathbf{y} = A\mathbf{x}$ that is simply a rotated version of $\mathbf{x}$ (by angle θ counterclockwise).

Solution:

To find the matrix representation of any operator, we need the basis of each space, and the effect of the operator on those basis vectors. We will use the standard basis vectors as shown in Figure 3.3.

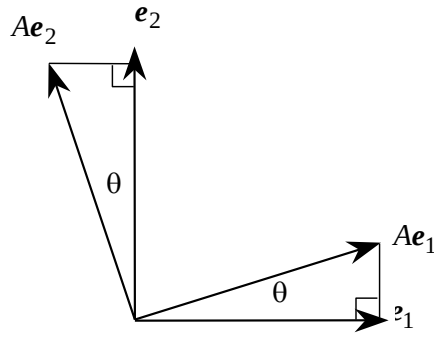


Figure 3.3 The effect of operator A on the basis vectors.

By decomposing the rotated vectors Ae_1 and Ae_2 along the original basis directions e_1 and e_2 , we can see using basic trigonometry that

$$\begin{aligned} Ae_1 &= \cos \theta \cdot e_1 + \sin \theta \cdot e_2 = [e_1 \mid e_2] \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \\ Ae_2 &= -\sin \theta \cdot e_1 + \cos \theta \cdot e_2 = [e_1 \mid e_2] \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix} \end{aligned} \quad (3.15)$$

Therefore, the matrix representation of A is

$$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Note that each column of this A -matrix is simply the corresponding representation in (3.15) above.

Applying this to the vector $\mathbf{x} = [1 \ 2]^T$, we get

$$A\mathbf{x} = \begin{bmatrix} \cos 30^\circ & -\sin 30^\circ \\ \sin 30^\circ & \cos 30^\circ \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -0.134 \\ 2.23 \end{bmatrix}$$

This rotation can be verified in Figure 3.4.

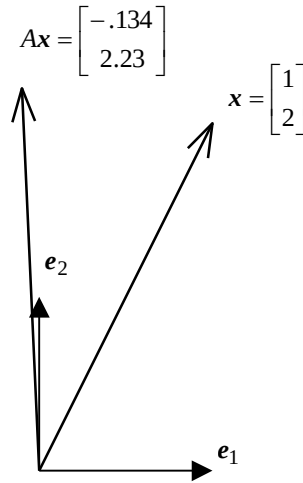


Figure 3.4 Pictorial representation of vector rotation.

Example 3.3: Composite Rotations: Roll, Pitch, and Yaw

Above we considered rotations in a plane. We can also consider rotations in three dimensions. However, in three dimensions, we must remember that rotations about three distinct axes are possible. The matrices that describe these rotations depend on which axis was used as the axis of rotation. The three possibilities, rotation about e_1 , e_2 , or e_3 , are depicted in Figure 3.5.

To achieve an arbitrary three-dimensional rotation, these rotations about coordinate axes may be applied in sequence, but such sequences would not be unique. If a three-dimensional orthonormal frame is arbitrarily oriented with respect to another three-dimensional orthonormal frame, there is generally an infinite number of different axis rotation steps that can be taken to arrive at one frame from the other.

Find the change of basis matrices for the three individual rotations and compute the composite rotation matrix that rotates a coordinate system by θ_R around e_1 , θ_P around e_2 , and θ_Y around e_3 , in that order.

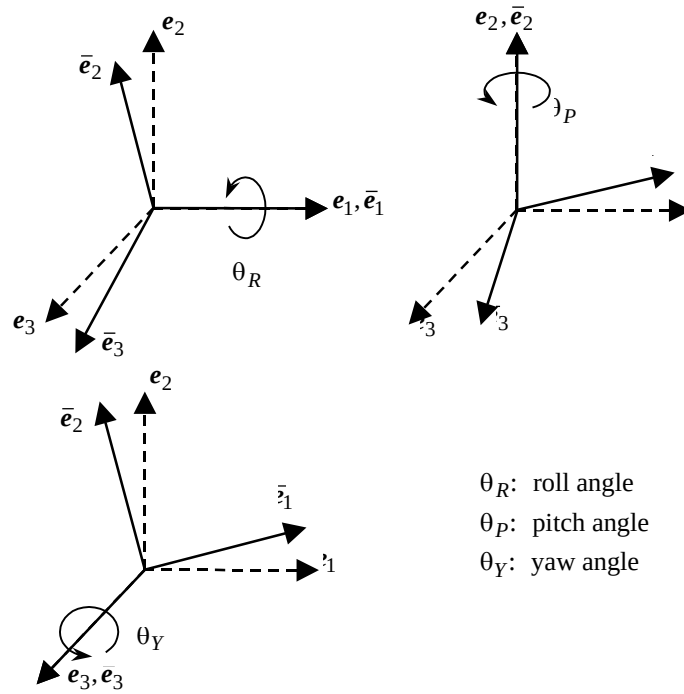


Figure 3.5 Axis rotations for pitch, roll, and yaw.

Solution:

The individual matrices may be computed by viewing the rotations in the plane in which they occur and decomposing the rotated axes in that plane, just as we did with the planar rotation example in Figure 3.3. With those same planar decompositions, the change of basis matrix, which can also be considered a rotation operator on a vector by Equation (2.33), for each rotation is:

$$R_R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_R & -\sin \theta_R \\ 0 & \sin \theta_R & \cos \theta_R \end{bmatrix} \quad R_P = \begin{bmatrix} \cos \theta_P & 0 & \sin \theta_P \\ 0 & 1 & 0 \\ -\sin \theta_P & 0 & \cos \theta_P \end{bmatrix}$$

and

$$R_Y = \begin{bmatrix} \cos \theta_Y & -\sin \theta_Y & 0 \\ \sin \theta_Y & \cos \theta_Y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

If a roll rotation is applied to a vector $\mathbf{x}$ first, followed by a pitch rotation, then a yaw rotation, the composite rotation can be written as the product

$$\begin{aligned} R &= R_Y R_P R_R \\ &= \begin{bmatrix} \cos \theta_Y \cos \theta_P & \cos \theta_Y \sin \theta_P \sin \theta_R - \sin \theta_Y \cos \theta_R & \dots \\ \sin \theta_Y \cos \theta_P & \sin \theta_Y \sin \theta_P \sin \theta_R + \cos \theta_Y \cos \theta_R & \dots \\ -\sin \theta_P & \cos \theta_P \sin \theta_R & \dots \end{bmatrix} \end{aligned} \quad (3.16)$$

$$\begin{bmatrix} \cos \theta_Y \sin \theta_P \cos \theta_R + \sin \theta_Y \sin \theta_R \\ \sin \theta_Y \sin \theta_P \cos \theta_R - \cos \theta_Y \sin \theta_R \\ \cos \theta_P \cos \theta_R \end{bmatrix}$$

Such rotations are useful for free-floating bodies, such as aircraft, missiles, spacecraft, and submarines, whose orientation is best described with respect to inertial coordinate frames. Rotations about body-centered coordinate axes are somewhat different (see a robotics text such as [5]).

Example 3.4: Matrix Representation of the Projection Operator

Let A be the orthogonal projection operator that projects vectors from $\mathfrak{R}^3$ into $\mathfrak{R}^2$. Suppose $\mathfrak{R}^2$ is given as the $\{\mathbf{e}_1, \mathbf{e}_2\}$ -plane, where the basis of $\mathfrak{R}^3$ is $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$. This situation is pictured in Figure 3.6. Find the matrix representation of the orthogonal projection operator $P: \mathfrak{R}^3 \rightarrow \mathfrak{R}^2$.

Solution:

To find the matrix representation of P , we again determine its effect on the basis vectors, two of which are reused in the new space $\mathfrak{R}^2$. In this case, we find that $P\mathbf{e}_1 = \mathbf{e}_1$, $P\mathbf{e}_2 = \mathbf{e}_2$, and $P\mathbf{e}_3 = \mathbf{0}$. Therefore the matrix of the projection operator is

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Of course this matrix must be (2×3) because the transformation takes vectors from three dimensions into two dimensions.

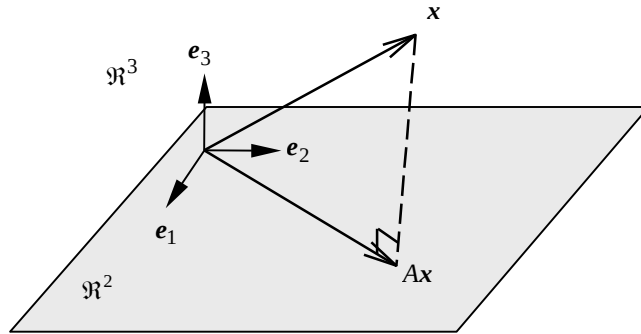


Figure 3.6 Orthogonal projection operation.

Example 3.5: Projection onto an Arbitrary Plane

Give the formula for the orthogonal projection of a vector x onto the plane that is orthogonal to a given vector n , as depicted in Figure 3.7 (unlike the example above, such a plane will, in general, not contain any of the coordinate axes).

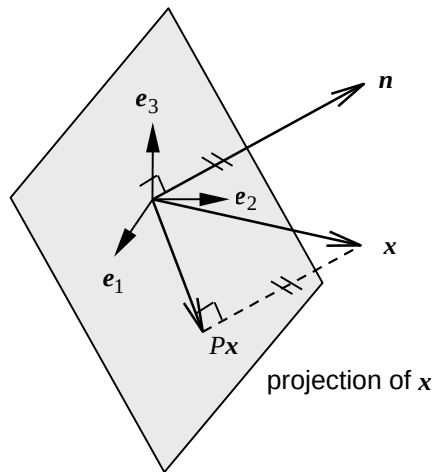


Figure 3.7 Orthogonal projection onto an arbitrary plane.

Solution:

To project $\mathbf{x}$ onto the plane normal to $\mathbf{n}$, we need to remove from $\mathbf{x}$ any of its components that lie along $\mathbf{n}$. To find such a component, we use an inner product. We then simply subtract this component. We must remember to normalize the vector $\mathbf{n}$ first: $\hat{\mathbf{n}} \triangleq \mathbf{n}/\|\mathbf{n}\|$. Then the length of the component of $\mathbf{x}$ along $\mathbf{n}$ will be $\hat{\mathbf{n}} \cdot \mathbf{x} = \hat{\mathbf{n}}^T \mathbf{x}$. Therefore, the projection is

$$P\mathbf{x} = \mathbf{x} - \hat{\mathbf{n}}(\hat{\mathbf{n}}^T \mathbf{x}) \quad (3.17)$$

or

$$P\mathbf{x} = (I - \hat{\mathbf{n}}\hat{\mathbf{n}}^T)\mathbf{x} \quad (3.18)$$

Example 3.6: Computer Graphics and Object Rotation

Consider a three-dimensional wire frame object that is to be drawn on a computer screen as it would be viewed by a person whose eye position is located by the vector $\mathbf{v}$. This eye position is subject to change, so a program is desired that can rotate the graphical representation of the object according to the viewer's location. Consider the object itself to be specified by a set of vectors $\mathbf{p}_i$, $i = 1, \dots, n$ that indicate the locations of the vertices. Assume that a graphics library is available that can draw lines between any pair of vertices provided, so that the user need only recompute the locations of these vertices in order to redraw the object. Formulate a set of operators that can be used to redraw the object.

Solution:

The same operator will be applied to each vertex vector, so we can consider working with any one of them, say $\mathbf{p}$. As depicted in Figure 3.8, the necessary transformation consists of two steps:

1. Rotation of the vertex vectors such that the eye position vector coincides with the z-axis of the viewing screen
2. Projection of the vertex vector onto the plane of the viewing screen

We could perform the projection onto the plane of the viewing screen first by using the results of the previous example. However, because the plane of the viewing screen is always aligned with the screen's z-axis, it will be simpler to first rotate the object. Afterward, the projection will simply be a projection onto a coordinate axis, and we can accomplish this by dropping its z-component.

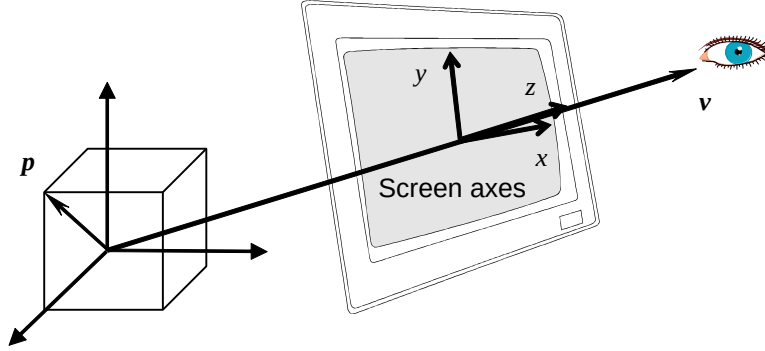


Figure 3.8 The rotation and projection of an object onto a viewing screen normal to eye position vector $\mathbf{v}$.

To do the rotation, we will find a rotation matrix that aligns $\mathbf{v}$ with the z -axis of the screen. From Figure 3.9 it is apparent that this operation can be accomplished in two steps: first a rotation by angle θ about the z -axis, followed by a rotation by angle ϕ about the x -axis. This is equivalent to a rotation using matrices R_Y and R_R with angles θ and ϕ , respectively. The composite rotation matrix is then

$$\begin{aligned}
 R &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \cos \phi \sin \theta & \cos \phi \cos \theta & -\sin \phi \\ \sin \phi \sin \theta & \sin \phi \cos \theta & 1 \end{bmatrix}
 \end{aligned} \tag{3.19}$$

The angles θ and ϕ can be found from

$$\theta = \tan^{-1} \frac{\hat{v}_1}{\hat{v}_2} \tag{3.20}$$

and

$$\phi = \tan^{-1} \frac{\sqrt{\hat{v}_1^2 + \hat{v}_2^2}}{\hat{v}_3} \tag{3.21}$$

where $\hat{v}_1$, $\hat{v}_2$, and $\hat{v}_3$ are the components of $\hat{\mathbf{v}} \triangleq \mathbf{v}/\|\mathbf{v}\|$.

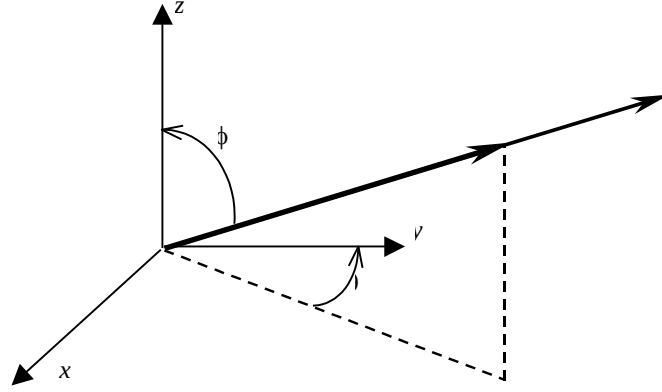


Figure 3.9 Rotation of the eye position to align with the screen z-axis.

After this alignment, the projection can be accomplished with the matrix operator

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad (3.22)$$

but this is equivalent to dropping the z-component of the rotated vector.

In summary, each vertex should be processed with the operators given in Equations (3.19) and (3.22):

$$(\mathbf{x}_i)_{\text{new}} = PR\mathbf{x}_i \quad (3.23)$$

after which the new two-dimensional vectors $(\mathbf{x}_i)_{\text{new}}$ are drawn and connected in screen coordinates.

3.1.3 Linear Operators in Different Bases

In the next chapter, we will be particularly concerned with transformations from a space into itself, such as $A: \mathfrak{R}^n \rightarrow \mathfrak{R}^n$. Of course, this will give a square $(n \times n)$ matrix, and it might simultaneously accomplish a change of basis. Let $\mathbf{X}_v$ denote such an n -dimensional space, equipped with basis $\{\mathbf{v}\}$. We say that the matrix A that transforms $\mathbf{X}_v \rightarrow \mathbf{X}_v$ is a *transformation* in the basis $\{\mathbf{v}\}$. This is for now a departure from our practice of referring to *vectors* in a basis, but since the representation of a vector will change as its basis does, then it should be expected

that the representation of an operator on that vector should change with the basis as well. The operator that transforms vectors expressed in a different basis $\{\hat{v}\}$ is therefore denoted by $\hat{A}$. Using this notation, we say that $y = Ax$ and $\hat{y} = \hat{A}\hat{x}$, where, clearly, $\hat{y}$ and $\hat{x}$ are vectors represented in basis $\{\hat{v}\}$.

The question of interest now is how we transform a matrix representation of an operator from one basis into another. We will do this using the change of basis matrices developed in the previous chapter. Using the change of basis matrix B as developed in Equation (2.33), we can write $\hat{y} = By$ and $\hat{x} = Bx$. Using these changes of basis, we have

$$\begin{aligned}\hat{y} &= \hat{A}\hat{x} \\ By &= \hat{A}Bx\end{aligned}$$

so

$$y = B^{-1}\hat{A}Bx$$

B^{-1} must exist because it is an $(n \times n)$ change of basis matrix and of full rank. Comparing this to the expression $y = Ax$, we see that

$$A = B^{-1}\hat{A}B \quad (3.24)$$

Equivalently, if we denote $M = B^{-1}$, then $A = M\hat{A}M^{-1}$, or

$$\hat{A} = M^{-1}AM \quad (3.25)$$

This change of basis for a matrix representation of a linear operator is known as a *similarity transformation*.^M It is said that A is *similar to* $\hat{A}$ and vice versa. It can be recognized that this transformation is exactly the process used in Chapter 1 when we first introduced state variables in different coordinates. What we were actually doing was changing the basis of the space of solutions to the differential equations.

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Example 3.7: Matrix Operators and Change of Basis

Consider the linear vector space $\mathbf{X}$ of polynomials in s of degree less than four, with real coefficients, over the field of reals. The operator $A: \mathbf{X} \rightarrow \mathbf{X}$ that takes such vectors $v(s)$ and returns the new vectors $w(s) = v''(s) + 2v'(s) + 3v(s)$ (where the prime denotes differentiation in s) is a linear operator (as is

differentiation in general). In this example, we shall find the matrix representation of this operator in two bases,

$$\{\mathbf{e}_i\} = \{s^3, s^2, s, 1\} \equiv \{\mathbf{e}_1(s), \mathbf{e}_2(s), \mathbf{e}_3(s), \mathbf{e}_4(s)\}$$

and

$$\{\bar{\mathbf{e}}_i\} = \{s^3 - s^2, s^2 - s, s - 1, 1\} \equiv \{\bar{\mathbf{e}}_1(s), \bar{\mathbf{e}}_2(s), \bar{\mathbf{e}}_3(s), \bar{\mathbf{e}}_4(s)\}$$

as well as the transformation between the two.

Solution:

First we will determine the effect of A on the basis vectors $\{\mathbf{e}_i\}$:

$$\begin{aligned} A\mathbf{e}_1 &= \mathbf{e}_1''(s) + 2\mathbf{e}_1'(s) + 3\mathbf{e}_1(s) \\ &= 6s + 6s^2 + 3s^3 \\ &= [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4] \begin{bmatrix} 3 \\ 6 \\ 6 \\ 0 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} A\mathbf{e}_2 &= \mathbf{e}_2''(s) + 2\mathbf{e}_2'(s) + 3\mathbf{e}_2(s) \\ &= 2 + 4s + 3s^2 \\ &= [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4] \begin{bmatrix} 0 \\ 3 \\ 4 \\ 2 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} A\mathbf{e}_3 &= \mathbf{e}_3''(s) + 2\mathbf{e}_3'(s) + 3\mathbf{e}_3(s) \\ &= 2 + 3s \\ &= [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4] \begin{bmatrix} 0 \\ 0 \\ 3 \\ 2 \end{bmatrix} \end{aligned}$$

$$\begin{aligned}
A\mathbf{e}_4 &= \mathbf{e}_4''(s) + 2\mathbf{e}_4'(s) + 3\mathbf{e}_4(s) \\
&= 3 \\
&= [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4] \begin{bmatrix} 0 \\ 0 \\ 0 \\ 3 \end{bmatrix}
\end{aligned} \tag{3.26}$$

Taking the four columns of coefficients in these four representations, we obtain the matrix

$$A = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 6 & 3 & 0 & 0 \\ 6 & 4 & 3 & 0 \\ 0 & 2 & 2 & 3 \end{bmatrix} \tag{3.27}$$

We can check this result by applying this transformation to the vector $\mathbf{v}(s) = s^2 + 1$, whose representation in $\{\mathbf{e}_i\}$ is $\mathbf{v} = [0 \quad 1 \quad 0 \quad 1]^T$. Multiplying this representation of $\mathbf{v}$ by the matrix A results in

$$A\mathbf{v} = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 6 & 3 & 0 & 0 \\ 6 & 4 & 3 & 0 \\ 0 & 2 & 2 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 4 \\ 5 \end{bmatrix} \tag{3.28}$$

Applying the transformation by direct differentiation gives

$$\mathbf{w}(s) = A\mathbf{v}(s) = \mathbf{v}''(s) + 2\mathbf{v}'(s) + 3\mathbf{v}(s) = 3s^2 + 4s + 5 \tag{3.29}$$

which is clearly the same vector as computed in (3.28) above.

How does this operator appear in the different basis $\{\bar{\mathbf{e}}_i\}$? We have two methods to determine the answer. We can either derive the matrix transformation directly in this basis, or we can perform a similarity transformation on the matrix A that was computed above in (3.27), as in (3.24). First, if we find the change of basis matrix between the two bases, we must compute the expansion

$$\mathbf{e}_j = \sum_{i=1}^n b_{ij} \bar{\mathbf{e}}_i$$

However, it is apparent by inspection that it is easier to compute the *inverse* relationship:

$$\begin{aligned} \bar{\mathbf{e}}_1 &= \mathbf{e}_1 - \mathbf{e}_2 = [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4] \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} \\ \bar{\mathbf{e}}_2 &= \mathbf{e}_2 - \mathbf{e}_3 = [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4] \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} \\ \bar{\mathbf{e}}_3 &= \mathbf{e}_3 - \mathbf{e}_4 = [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4] \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} \\ \bar{\mathbf{e}}_4 &= \mathbf{e}_4 = [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4] \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \end{aligned}$$

Because this is the inverse relationship, the matrix we arrive at is the inverse matrix, $M = B^{-1}$. By gathering the coefficient columns, this matrix is

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

from which we can compute

$$B = M^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Now using the formula $A = B^{-1}\bar{A}B$ from Equation (3.24), we find that

$$\bar{A} = BAB^{-1} = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 6 & 3 & 0 & 0 \\ 8 & 4 & 3 & 0 \\ 6 & 4 & 2 & 3 \end{bmatrix}$$

Now to check this result by finding $\bar{A}$ in another way, we can find the effect of the original operator on vectors that have already been expressed in the $\{\bar{\mathbf{e}}_i\}$ basis. First, we should find the expression for $\bar{\mathbf{v}}(s)$, i.e., the same vector $\mathbf{v}$, but in the “bar” basis:

$$\bar{\mathbf{v}} = B\mathbf{v} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix}$$

which is simply the matrix-vector notation for

$$1 \cdot \bar{\mathbf{e}}_2(s) + 1 \cdot \bar{\mathbf{e}}_3(s) + 2 \cdot \bar{\mathbf{e}}_4(s) = (s^2 - s) + (s - 1) + 2 = s^2 + 1 = \mathbf{v}(s)$$

Now applying the matrix operation in the $\{\bar{\mathbf{e}}_i\}$ basis,

$$\bar{A}\bar{\mathbf{v}} = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 6 & 3 & 0 & 0 \\ 8 & 4 & 3 & 0 \\ 6 & 4 & 2 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 7 \\ 12 \end{bmatrix}$$

Writing this representation out explicitly in terms of the “barred” basis vectors,

$$\bar{A}\bar{\mathbf{v}}(s) = 3\bar{\mathbf{e}}_2 + 7\bar{\mathbf{e}}_3 + 12\bar{\mathbf{e}}_4 = 3(s^2 - s) + 7(s - 1) + 12(1) = 3s^2 + 4s + 5$$

which of course is exactly as we expect: the same vector we computed in Equation (3.28). In other terms, we have demonstrated the commutativity of the Figure 3.10. So-called “commutative diagrams” show how different paths, each representing an operation or transformation taken in order, achieve the same result. To get from one vector to another, one performs the operation indicated next to the arrow, in the direction indicated. If a path must be traversed in reverse (“against” the arrow), then the inverse transformation is used. One can use this diagram to illustrate that $\bar{A} = BAB^{-1}$.

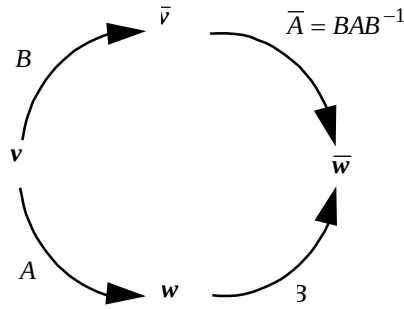


Figure 3.10 Commutative diagram showing different paths to the transformed vector $\bar{w}$.

3.2 Operators as Spaces

It can be shown that the set of all linear operators from one space into another, or from a space into itself, is itself a linear vector space. Because we have been representing linear operators as matrices, the space of operators is, in fact, a space of matrices. It is also an *algebra*, as defined in the previous chapter, because it has a multiplication operation, which we have been using all along whenever we multiplied two compatible matrices.

Because the set of operators is a space, it must satisfy all the other requirements of linear vector spaces, as presented in Chapter 2. For example, operators have an addition operation, which takes the form of the familiar element-by-element matrix addition, where

$$A_1 \mathbf{x} + A_2 \mathbf{x} = (A_1 + A_2) \mathbf{x}$$

3.2.1 Operator Norms

Operators also have norms. An operator norm,^M like a vector norm, is intuitively a measure of how big the operator is, and how “big” it is depends on what it does when it operates on a vector. In this sense, a norm describes an operator’s maximal “magnification” properties. With this interpretation, we state a general norm for an operator (i.e., a matrix) as

$\text{norm}(A)$

$$\|A\| = \sup_{\mathbf{x} \neq 0} \frac{\|A\mathbf{x}\|}{\|\mathbf{x}\|} = \sup_{\|\mathbf{x}\|=1} \|A\mathbf{x}\| \quad (3.30)$$

where *sup* (“supremum”) indicates the least upper bound. Of course, as we have seen, there are many ways to define the norm $\|\mathbf{x}\|$ of $\mathbf{x}$. Consequently, there are many different operator norms. However, all operator norms must satisfy the following properties:

$$1. \quad \|A\mathbf{x}\| \leq \|A\| \cdot \|\mathbf{x}\| \text{ for all vectors } \mathbf{x} \quad (3.31)$$

$$2. \quad \|A_1 + A_2\| \leq \|A_1\| + \|A_2\| \quad (3.32)$$

$$3. \quad \|A_1 A_2\| \leq \|A_1\| \cdot \|A_2\| \quad (3.33)$$

$$4. \quad \|\alpha A\| = |\alpha| \|A\| \quad (3.34)$$

Corresponding to the vector norm definitions given in Chapter 2, the following matrix norms may be used:

$$1. \quad \|A\|_1 = \max_j \left(\sum_{i=1}^n |a_{ij}| \right) \quad (3.35)$$

$$2. \quad \|A\|_2 = \left[\max_{\|\mathbf{x}\|=1} \{ \bar{\mathbf{x}}^T A^T A \mathbf{x} \} \right]^{1/2} \quad (3.36)$$

where the overbar denotes complex conjugate. This is the euclidean matrix norm. Note that this definition does not indicate how this norm is actually computed. For that we will appeal to the definitions of eigenvalues and singular values in the next chapter.

$$3. \quad \|A\|_\infty = \max_i \left(\sum_{j=1}^n |a_{ij}| \right) \quad (3.37)$$

$$4. \quad \|A\|_F = \left[\text{tr}(\bar{A}^T A) \right]^{1/2} = \left[\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2 \right]^{1/2} \quad (3.38)$$

trace (A)

where $\text{tr}(\cdot)$ denotes the trace^M of the matrix argument, which is the sum of the elements on the diagonal. This is called the *Frobenius norm*.

Regardless of which norm we choose, we say that the linear operator A is *bounded* if there exists a finite constant κ such that $\|A\mathbf{x}\| \leq \kappa\|\mathbf{x}\|$ for all vectors $\mathbf{x}$ in the space.

' (single
apostrophe
denotes
transpose)

3.2.2 Adjoint Operators

We have used the transpose^M of several matrices throughout the definitions of such quantities as inner products and norms. The transpose, however, is a special kind of operator, which satisfies the following definition:

Adjoint Operator: The *adjoint* of linear operator A is denoted A^* and must satisfy the relationship

$$\langle A\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, A^*\mathbf{y} \rangle \quad (3.39)$$

for all vectors $\mathbf{x}$ and $\mathbf{y}$.

The operator A^* may itself be represented by a matrix. For matrix representations of finite-dimensional operators, A^* is equal to the complex conjugate transpose of matrix A , $A^* = \overline{A}^T$ in an orthonormal basis. If A is real, therefore, the representation of A^* , in an orthonormal basis, is simply equal to the matrix A^T .

Sometimes the adjoint of an operator (matrix) is the same as the operator (matrix) itself, in which case it is called a *hermitian* operator (matrix), and the operator is referred to as *self-adjoint*. Note that if the basis in which an operator is represented is not orthonormal, then a symmetric matrix representation does *not* imply a self-adjoint operator. If a hermitian matrix is actually real, then it is *symmetric*, and $A = A^T$.

3.3 Simultaneous Linear Equations

The solution of systems of simultaneous linear algebraic equations is an important application of spaces and operator theory. Undoubtedly the reader has had some exposure to systems of simultaneous linear equations before, and has some experience solving such systems using methods such as gaussian elimination or Cramer's rule. It is not our purpose to discuss such numerical computations but rather to consider the solution of simultaneous algebraic equations from a vector space viewpoint. This will provide the geometric insight needed to explain certain phenomena observed in engineering, such as the existence of multiple solutions to a problem or the nonexistence of solutions. At the end of this section, we will

present examples that will lead to important conclusions about properties of systems that are not discussed until later in the book, e.g., controllability, observability, and estimation.

To begin, consider a set of m equations in terms of n unknown quantities, $x_1, \dots, x_n$:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= y_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= y_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= y_m \end{aligned} \quad (3.40)$$

By writing this set of equations in the form

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$$

we can express this relationship as $A\mathbf{x} = \mathbf{y}$, where $\mathbf{x} \in \mathfrak{R}^n$, and $\mathbf{y} \in \mathfrak{R}^m$, and the matrix A is considered to be a linear operator $A: \mathfrak{R}^n \rightarrow \mathfrak{R}^m$.

As we have said, we wish to investigate the circumstances of when, if, and how many solutions exist for this system of equations. By interpreting the system as a linear operator, we can rephrase the question, “Is the given vector $\mathbf{y}$ in the range space $R(A)$?” We can use yet another notation to clarify this interpretation of the problem. Suppose the i^{th} column of the matrix A is considered to be a vector $\mathbf{a}_i$ from the space of m -tuples $\mathfrak{R}^m$. Then, the above notation becomes

$$A\mathbf{x} = \begin{bmatrix} \mathbf{a}_1 \end{bmatrix} x_1 + \begin{bmatrix} \mathbf{a}_2 \end{bmatrix} x_2 + \cdots + \begin{bmatrix} \mathbf{a}_n \end{bmatrix} x_n = \begin{bmatrix} \mathbf{y} \end{bmatrix}$$

For equality to hold, then, the vector $\mathbf{y}$ would have to be a linear combination of the columns of A . If this is the case, then $\mathbf{y}$ will be linearly dependent on the columns of A . The problem thus reduces to a test of linear independence of vectors.

We can perform this test of independence by forming the new matrix $W = [A \mid \mathbf{y}]$ by appending^M $\mathbf{y}$ onto A as its rightmost column. If indeed $\mathbf{y}$ is

:(colon denotes
matrix append)

dependent on the columns of A , then we should have $r(A) = r(W)$, because $\mathbf{y}$ will not have added any new “directions” to the space already spanned by the columns of A . Then, of course, $\mathbf{y} \in R(A)$. Otherwise, i.e., if $\mathbf{y}$ were linearly independent of the columns of A , then $r(W)$ would be equal to $r(A) + 1$. In summary, then, we can say that

$$r(A) = r(W) \quad \text{if at least one solution exists to } A\mathbf{x} = \mathbf{y}$$

$$r(A) \neq r(W) \quad \text{if there are no solutions to } A\mathbf{x} = \mathbf{y}$$

3.3.1 Multiple and Nonexistent Solutions

Note that in the first statement above, we have emphasized the fact that if $r(A) = r(W)$, then there is not only a solution, but there may be more than one solution. In fact, if $r(A) = r(W)$, there is either a unique solution or else there are an infinite number of solutions. This situation depends on the relationship between $r(A) [= r(W)]$ and the dimension of the space $\mathcal{R}^n$, which of course is n .

First, if $r(A) = n$, then the solution to the system $A\mathbf{x} = \mathbf{y}$ will be *unique*. This is because there are exactly n columns in the matrix A , and if $r(A) = n$, then all the columns are linearly independent. This set of n columns therefore forms a basis for $\mathcal{R}^n$. Knowing that $\mathbf{y} \in R(A)$, then we know from Chapter 2 that the representation of $\mathbf{y}$ in this basis is unique. The representation will be exactly the set of coefficients $\{x_i\}$ that we are seeking.

If, on the other hand, $r(A) < n$, then there are fewer than n columns of A that are linearly independent, i.e., the columns of A form a linearly dependent set. The set $\{\mathbf{a}_i\}$ still spans the subspace in which $\mathbf{y}$ lies, because $r(A) = r(W)$, but because there are too many $\mathbf{a}_i$'s to form a basis, the representation of $\mathbf{y}$ in the set $\{\mathbf{a}_i\}$ is not unique.

If the solution is not unique, one may ask why the set of possible solutions is infinite and not finite. To answer this question, recall that $q(A) = n - r(A)$. If we have $r(A) < n$, then our result is that $q(A) > 0$. The significance of this is that the nullity is the same as the dimension of the null space of a matrix. Therefore, $q(A) > 0$ implies that the matrix (operator) A has a nontrivial null space. That is, there exists at least one nonzero vector $\mathbf{x}_0$ such that $A\mathbf{x}_0 = 0$. This being true, we can always multiply $\mathbf{x}_0$ by an arbitrary non-zero scalar, and the result will still be in the null space of A . We are then allowed to add to any solution of $A\mathbf{x} = \mathbf{y}$ such a vector in $N(A)$, and a different solution will result. That is,

$$A(\mathbf{x} + \alpha \mathbf{x}_0) = A\mathbf{x} + \alpha A\mathbf{x}_0 = A\mathbf{x} + 0 = A\mathbf{x} = \mathbf{y}$$

In general, we can think of solutions of algebraic systems of equations in a way similar to the solutions of nonhomogeneous differential equations: Each complete solution $\mathbf{x}$ consists of two parts: the *particular part* $\mathbf{x}_p$ for which $A\mathbf{x}_p = \mathbf{y}$ and the *homogeneous part* for which $A\mathbf{x}_h = 0$. The general solution is of the form $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$. Often, the nullity of A is even greater than one, so that there is an entire multidimensional subspace $N(A)$ from which homogeneous terms can be found.

Finding the Null Space

In practice, when solving such a set of linear algebraic equations, the null space of the matrix A can be easily determined by computer^M using an analysis package such as MATLAB. However, it is useful to learn the “manual” procedure for finding this null space and for solving the equation $A\mathbf{x} = \mathbf{y}$ in general.^M

To begin, we will determine the row-reduced echelon form of the matrix A . The *row-reduced echelon form*^M is a simplification of the matrix through a series of so-called “elementary row operations,” which simplify the matrix but leave the null space intact. The elementary row operations that are allowed are:

1. Multiply any row by a nonzero scalar.
2. Interchange any two rows.
3. Add a multiple of one row to another row.

The goal of performing these operations is first to reduce the matrix to an upper triangular matrix, forcing zeros to appear in the portion of the matrix below the main diagonal. (The diagonal of a nonsquare matrix is simply the sequence of elements, $a_{11}, a_{22}, \dots$). After this is performed, a matrix of the following shape will result:

$$\begin{bmatrix} 0 & \boxed{\times} & \times & \times & \times & \times \\ 0 & 0 & 0 & \boxed{\times} & \times & \times \\ 0 & 0 & 0 & 0 & \boxed{\times} & \times \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

In this matrix, the boxes indicate the first nonzero element in each row. This is called the *pivot* element, and the number of pivot elements (in this case, three) is the rank of the matrix. This matrix has the same null space as the original matrix, but this null space is easier to determine. (In fact, each subset of rows in the row-reduced echelon form spans the same space as the corresponding rows did

`null(A)`

`\(backslash)`

`rref(A)`

originally, but if row interchange operations are used, the columns of the reduced matrix do not necessarily span the same “column space.”) Hence, if this matrix were the A -matrix in a problem $Ax = y$, then $n = 6$, and the dimension of the null space would be $q(A) = n - r(A) = 6 - 3 = 3$. Because the pivot elements occur in columns two, four, and five, the variables x_2 , x_4 , and x_5 will be called *pivot* variables. The other variables, x_1 , x_3 , and x_6 , are referred to as *free* variables.

$\text{null}(A)$

To generate a basis^M for the $N(A)$, we generate a linearly independent set of free variable vectors and solve the system of row-reduced equations for the pivot variables. The easiest way to generate this independent set is to choose one free variable as a ‘1,’ the rest ‘0’s, and solve for the pivot variables. This set of variables is a basis vector for $N(A)$. We then set the next free variable equal to ‘1,’ the others to ‘0,’ and again solve, getting another basis vector for the null space. This process continues until we get $q(A)$ such vectors, after which we have found the basis for $N(A)$. In common terminology, we will have found $N(A)$ itself.

It is helpful to continue this discussion with the aid of an example. Suppose we are interested in the particular and homogeneous solutions of the equation

$$\begin{bmatrix} 1 & -1 & 2 \\ -1 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$$

We can find the ranks of matrices A and W by constructing W and row reducing it first:

$$W = [A \mid y] = \left[\begin{array}{ccc|c} 1 & -1 & 2 & 8 \\ -1 & 2 & 0 & 2 \end{array} \right]$$

The process of row reduction is fairly easy in this example. If we replace row 2 by the sum of row 1 and row 2, we arrive at the reduced form:

$$W' = \left[\begin{array}{ccc|c} \boxed{1} & -1 & 2 & 8 \\ 0 & \boxed{1} & 2 & 10 \end{array} \right] \quad (3.41)$$

We immediately see that there are two pivot elements (two nonzero rows), and hence, $r(W) = 2$. It is also apparent that the result would be the same if we disregard the right-most column, giving $r(A) = 2$ as well. Therefore, $r(A) = r(W) = 2$, and there must be at least one solution. Furthermore, $n = 3$, so $r(A) < n$, and there will be an infinite number of solutions, where

$\dim(N(A)) = q(A) = n - r(A) = 3 - 2 = 1$. The free variable is recognized as x_3 , so to generate a basis for the null space, consider the homogeneous situation

$$\begin{bmatrix} 1 & -1 & 2 & | & 0 \\ 0 & 1 & 2 & | & 0 \end{bmatrix} \quad (3.42)$$

or think of this matrix as the set of equations

$$\begin{aligned} x_1 + (-1)x_2 + 2x_3 &= 0 \\ x_2 + 2x_3 &= 0 \end{aligned}$$

Set $x_3 = 1$, then solve for the pivot variables: $x_2 = -2$, and $x_1 = -4$. Thus, the vector $\mathbf{x}_0 = [-4 \ -2 \ 1]^T$ is a basis vector for $N(A)$, as is any scalar multiple of $\mathbf{x}_0$.

To complete the problem, we must now find the particular part of the general solution. Toward this end, we can pursue the row reduction of the W matrix further. Consider taking (3.41) and performing additional elementary row operations until all elements above any pivot elements become zero. This is the “complete” row reduction and can also be used to determine the existence of any solutions. Specifically for (3.41), we replace row 1 by the sum of row 1 and row 2. This gives

$$W'' = \begin{bmatrix} 1 & 0 & 4 & | & 18 \\ 0 & 1 & 2 & | & 10 \end{bmatrix}$$

Because x_3 is a free variable, a solution to this equation, rewritten as

$$\begin{aligned} x_1 + 4x_3 &= 18 \\ x_2 + 2x_3 &= 10 \end{aligned}$$

is

$$\begin{aligned} x_1 &= 18 - 4x_3 \\ x_2 &= 10 - 2x_3 \end{aligned}$$

From this expression, we see that the homogeneous solution is “shifted” in space by a vector $\mathbf{x}_p = [18 \ 10 \ 0]^T$, and this is the particular solution to the original problem, i.e.,

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_0 = \begin{bmatrix} 18 \\ 10 \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} -4 \\ -2 \\ 1 \end{bmatrix}$$

Two further examples are provided below to further illustrate the possible situations:

Example 3.8: Unique and Nonexistent Solutions

Find every solution to the systems of equations, if they exist:

$$a) \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 3 & 2 & 1 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 4 \\ 7 \\ 1 \end{bmatrix} \quad \text{and} \quad b) \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 2 \\ 3 \\ -4 \end{bmatrix}$$

Solution:

a) We construct

$$W = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 2 & 1 & 2 & 7 \\ 3 & 2 & 1 & 1 \end{array} \right]$$

for which complete row reduction (including elimination of nonzero values above pivot variables) produces

$$W'' = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2.125 \\ 0 & 1 & 0 & -4.5 \\ 0 & 0 & 1 & 3.625 \end{array} \right]$$

Because there are three pivot rows in both A'' and W'' , then $r(A) = r(W) = 3 = n$, implying that there is a unique solution. This solution is apparent in the complete row reduction as $\mathbf{x} = [2.125 \quad -4.5 \quad 3.625]^T$.

b) We have a complete row reduction of

$$W'' = \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$$

We can see that the A'' portion (to the left of the partition) of this matrix has two pivot rows, and hence $r(A) = 2$, while the whole matrix has three pivot rows, so $r(W) = 3$. Thus $r(A) \neq r(W)$, and there is no solution to this problem.

In the special case that A is square ($n \times n$) and $r(A) = r(W) = n$, then the solution is usually found with the much simpler technique $\mathbf{x} = A^{-1}\mathbf{y}$. This is possible, of course, because the full rank of the square A -matrix implies that its inverse exists. In the case that A is not square, we must consider two possible cases: either $m < n$ or $m > n$. These situations are addressed next.

More Unknowns Than Equations

In the case that $m < n$, we have more unknown elements of the vector $\mathbf{x}$ than we have equations to determine these unknowns. This is known as an *underdetermined* system and requires some additional analysis. When $m < n$, there is no unique solution, because we know that $r(A) \leq \min(m, n)$ and $(\min(m, n) = m) < n$. Thus, when solving for one solution to $A\mathbf{x} = \mathbf{y}$, we have the option of determining which of the infinitely many solutions to determine. We will seek the solution with the minimum norm from among all possibilities. Minimum-norm solutions are often sought when vector $\mathbf{x}$ represents a physical quantity with a cost associated with it, such as a control input, kinetic energy, or fuel consumption.

We will actually minimize one-half of the *squared* norm of $\mathbf{x}$, which will have a minimum at the same values as the norm itself. To do this, we introduce the concept of constrained minimization and cast the problem as follows: Find, among all $\mathbf{x}$, the value of $\mathbf{x}$ that minimizes $\frac{1}{2} \mathbf{x}^T \mathbf{x}$ subject to the constraint that $A\mathbf{x} = \mathbf{y}$, or, equivalently, $A\mathbf{x} - \mathbf{y} = \mathbf{0}$. To minimize a function subject to an equality constraint, a variable called a *LaGrange multiplier* is used. Using the LaGrange multiplier γ , a function called a hamiltonian is formed:

$$H = \frac{1}{2} \mathbf{x}^T \mathbf{x} + \gamma^T (A\mathbf{x} - \mathbf{y})$$

The $(m \times 1)$ vector LaGrange multiplier serves to create a scalar secondary optimization criterion from the vector expression $A\mathbf{x} - \mathbf{y}$. Hence, the hamiltonian is a scalar quantity that we wish to minimize over all possible $\mathbf{x}$ and γ (because γ is as yet undetermined). Therefore, because we are minimizing over two variables, we must use two derivatives and set each equal to zero:

$$\frac{\partial^T H}{\partial \mathbf{x}} = \mathbf{x} + A^T \gamma = 0 \quad (3.43)$$

and

$$\frac{\partial^T H}{\partial \gamma} = A\mathbf{x} - \mathbf{y} \equiv 0 \quad (3.44)$$

Note that the second of these partial derivatives returns exactly the original constraint equation. This is expected, of course, since that constraint must always be maintained. Rearranging (3.43) and multiplying both sides by the quantity A^T , we get

$$A\mathbf{x} = -AA^T\gamma$$

However, Equation (3.44) provides that $A\mathbf{x} = \mathbf{y}$, so we have

$$\mathbf{y} = -AA^T\gamma$$

or

$$\gamma = -(AA^T)^{-1}\mathbf{y} \quad (3.45)$$

Substituting this expression back into (3.43), we have the solution

$$\mathbf{x} = A^T(AA^T)^{-1}\mathbf{y} \quad (3.46)$$

This is the “shortest” vector $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{y}$.

We should note that in (3.46), the inverse of the matrix product (AA^T) will only be guaranteed in the event that A is of full rank—in this case, if $r(A) = m$. In the event that $r(A) < m$, then some of the original equations to be solved are linearly dependent on the others and are therefore redundant. One can rectify that situation by simply discarding the linearly dependent equations, one at a time, until $r(A) = m$.

`pinv(A)`

The quantity $A^T(AA^T)^{-1}$ in (3.44) is called a *pseudoinverse*^M of A (specifically, the unweighted Moore-Penrose pseudoinverse), denoted A^+ . Pseudoinverses are special classes of generalized inverses and are discussed in more detail in Appendix A. Pseudoinverses are different from true inverses in that they are not unique, so we should keep in mind that they are providing only one of possibly many solutions. For example, for any nonsingular symmetric matrix B , the pseudoinverse solution

$$\mathbf{x} = A_B^+ \mathbf{y} \triangleq B A^T (A B A^T)^{-1} \mathbf{y}$$

will also satisfy the equation $A\mathbf{x} = \mathbf{y}$.

More Equations Than Unknowns

We will now consider the situation in which $m > n$. This occurs when there are more equations than unknowns, which is referred to as an *overdetermined* system. This is the more unusual of the two situations because in order for a solution to exist, we are asking that the vector $\mathbf{y}$, which has relatively many components, be a linear combination of relatively few columns of the A -matrix. Nevertheless, if we find that $r(A) = r(W)$, then a solution must exist. We will approach the search for a solution differently from the underdetermined case, but we will find a solution with similar qualities.

In this case, we will define an “error” signal:

$$\mathbf{e} = \mathbf{y} - A\mathbf{x} \quad (3.47)$$

Now we will seek a solution to the problem of minimizing one-half the squared-norm of $\mathbf{e}$ over all possible vectors $\mathbf{x}$, i.e., find $\mathbf{x}$ to minimize $\frac{1}{2} \mathbf{e}^T \mathbf{e}$. If indeed a solution exists, then this error should, of course, be zero.

To perform the minimization, we first substitute in the definition of the error, then simplify:

$$\begin{aligned} \frac{1}{2} \mathbf{e}^T \mathbf{e} &= \frac{1}{2} (\mathbf{y} - A\mathbf{x})^T (\mathbf{y} - A\mathbf{x}) \\ &= \frac{1}{2} (\mathbf{y}^T - \mathbf{x}^T A^T) (\mathbf{y} - A\mathbf{x}) \\ &= \frac{1}{2} (\mathbf{y}^T \mathbf{y} - \mathbf{x}^T A^T \mathbf{y} - \mathbf{y}^T A \mathbf{x} + \mathbf{x}^T A^T A \mathbf{x}) \end{aligned} \quad (3.48)$$

The two middle terms in the parentheses above are transposes of each other, but they are also scalars. They are, therefore, equal. Therefore

$$\frac{1}{2} \mathbf{e}^T \mathbf{e} = \frac{1}{2} (\mathbf{y}^T \mathbf{y} - 2\mathbf{x}^T A^T \mathbf{y} + \mathbf{x}^T A^T A \mathbf{x})$$

Taking a derivative of this and setting equal to zero,

$$\frac{\partial}{\partial \mathbf{x}} \left(\frac{1}{2} \mathbf{e}^T \mathbf{e} \right) = \frac{1}{2} (-2A^T \mathbf{y} + 2A^T A \mathbf{x}) = 0$$

so

$$A^T \mathbf{y} = A^T A \mathbf{x}$$

or

$$\mathbf{x} = (A^T A)^{-1} A^T \mathbf{y} \quad (3.49)$$

One can easily notice the similarity between this result and (3.46). In fact, the expression $(A^T A)^{-1} A^T$ is another type of pseudoinverse.

Example 3.9: Pseudoinverse Solution

Find the solution to

$$\begin{bmatrix} 2 & 2 \\ 1 & 2 \\ 1 & 0 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix}$$

Solution:

One can check that $r(A) = r(W)$, which is also obvious by inspection, since the vector $\mathbf{y}$ is easily seen to be the sum of the two columns of A . However, using the pseudoinverse as an exercise, we obtain

$$\mathbf{x} = (A^T A)^{-1} A^T \mathbf{y} = \begin{bmatrix} 1/3 & -1/3 & 2/3 \\ 0 & 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

To further demonstrate that this is an exact solution, we can compute the error:

$$\mathbf{e} = \mathbf{y} - A\mathbf{x} = \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 & 2 \\ 1 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

which, of course, has zero norm.

Example 3.10: Controllability of Discrete-Time Systems

In this example we will consider a discrete-time linear system described by the state equations

$$\begin{aligned} \mathbf{x}(k+1) &= A\mathbf{x}(k) + Bu(k) \\ y(k) &= C\mathbf{x}(k) + Du(k) \end{aligned} \quad (3.50)$$

where $\mathbf{x}(k)$ denotes an n -dimensional state vector for a system at time step k and $u(k)$ is a scalar input term. Under what conditions will it be possible to force the state vector to be zero, $\mathbf{x}(k) = 0$, for some k , by applying a particular sequence of inputs $u(0), u(1), \dots, u(k-1)$? The result will be important for the analysis of controllability in Chapter 8.

Solution:

We will first “execute” a few steps in the recursion of this equation:

At time $k = 0$:

$$\mathbf{x}(1) = A\mathbf{x}(0) + Bu(0) \quad (3.51)$$

At time $k = 1$:

$$\begin{aligned} \mathbf{x}(2) &= A\mathbf{x}(1) + Bu(1) \\ &= A[A\mathbf{x}(0) + Bu(0)] + Bu(1) \\ &= A^2\mathbf{x}(0) + ABu(0) + Bu(1) \end{aligned} \quad (3.52)$$

At time $k = 2$:

$$\begin{aligned} \mathbf{x}(3) &= A\mathbf{x}(2) + Bu(2) \\ &= A[A^2\mathbf{x}(0) + ABu(0) + Bu(1)] + Bu(2) \\ &= A^3\mathbf{x}(0) + A^2Bu(0) + ABu(1) + Bu(2) \end{aligned} \quad (3.53)$$

We can see that at any give time k , in order to get $\mathbf{x}(k) = 0$, we will have

$$\begin{aligned} 0 &= \mathbf{x}(k) \\ &= A^k\mathbf{x}(0) + A^{k-1}Bu(0) + A^{k-2}Bu(1) + \dots + ABu(k-2) + Bu(k-1) \end{aligned} \quad (3.54)$$

This can be rearranged as

$$-A^k \mathbf{x}(0) = \begin{bmatrix} B & \cdots & A^{k-2}B & A^{k-1}B \end{bmatrix} \begin{bmatrix} u(k-1) \\ \vdots \\ u(1) \\ u(0) \end{bmatrix} \quad (3.55)$$

If we introduce the notation

$$P \triangleq \begin{bmatrix} B & \cdots & A^{k-2}B & A^{k-1}B \end{bmatrix}$$

and

$$\mathbf{u} \triangleq \begin{bmatrix} u(k-1) \\ \vdots \\ u(1) \\ u(0) \end{bmatrix}$$

then we have

$$-A^k \mathbf{x}(0) = P\mathbf{u}$$

In this expression, we note that $-A^k \mathbf{x}(0) \in \Re^n$, $P \in \Re^{n \times k}$, and $\mathbf{u} \in \Re^k$. This is now in the form of a simultaneous system of linear algebraic equations to solve. If $-A^k \mathbf{x}(0) \in R(P)$, there exists a solution. Because we are allowing arbitrary values for both A and $\mathbf{x}(0)$, the only way to guarantee the existence condition is for $R(P) = \Re^n$ or, equivalently, for $r(P) = n$. Because the P -matrix is $n \times k$, this will require that $k \geq n$, i.e., we must wait for at least n samples to be computed, but in no case will $r(P) > n$. Note that we do not necessarily require that the solution $\mathbf{u}$ be unique, only that at least one solution exists.

A system that satisfies this criterion is said to be *controllable*. (Though there is more to controllability than just this. See Chapter 8.)

Example 3.11: Observability of Discrete-Time Systems

In this example, we look at a “dual” problem. Instead of finding an input that results in a particular state, we wish to determine an initial state given a sequence of inputs and outputs. Suppose after k iterations of Equation (3.50), we have recorded the sequence of applied inputs $u(k), \dots, u(0)$, and the measured sequence of responses $y(k), \dots, y(0)$. Under what conditions will it be possible to determine the value of the unknown initial state vector $\mathbf{x}(0)$?

Solution:

As we did in the previous example, we begin by listing several steps (up to $k-1$) in the recursive execution of the state equations in (3.50).

$$\begin{aligned}
 y(0) &= Cx(0) + Du(0) \\
 y(1) &= Cx(1) + Du(1) \\
 &= C[Ax(0) + Bu(0)] + Du(1) \\
 &= CAx(0) + CBu(0) + Du(1) \\
 \\
 y(2) &= Cx(2) + Du(2) \\
 &= C[A^2x(0) + ABu(0) + Bu(1)] + Du(2) \\
 &= CA^2x(0) + CABu(0) + CBu(1) + Du(2) \\
 &\vdots \\
 y(k-1) &= CA^{k-1}x(0) + CA^{k-2}Bu(0) + \cdots + CBu(k-2) + Du(k-1)
 \end{aligned}$$

Rearranging these equations into vector-matrix form,

$$\begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ \vdots \\ y(k-1) \end{bmatrix} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{k-1} \end{bmatrix} x(0) + \begin{bmatrix} D & 0 & 0 & \cdots & 0 \\ CB & D & 0 & \cdots & 0 \\ CAB & CB & D & \cdots & 0 \\ \vdots & & & \ddots & \\ CA^{k-2}B & \cdots & & & D \end{bmatrix} \begin{bmatrix} u(0) \\ u(1) \\ u(2) \\ \vdots \\ u(k-1) \end{bmatrix}$$

The term containing the large matrix and the input terms $u(\cdot)$ can be moved to the left side of the equal sign and combined with the vector containing the y 's to give a new vector called Ψ_{k-1} :

$$\Psi_{k-1} \triangleq \begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ \vdots \\ y(k-1) \end{bmatrix} - \begin{bmatrix} D & 0 & 0 & \cdots & 0 \\ CB & D & 0 & \cdots & 0 \\ CAB & CB & D & \cdots & 0 \\ \vdots & & & \ddots & \\ CA^{k-2}B & \cdots & & & D \end{bmatrix} \begin{bmatrix} u(0) \\ u(1) \\ u(2) \\ \vdots \\ u(k-1) \end{bmatrix} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{k-1} \end{bmatrix} x(0)$$

(3.56)

Let

$$Q \triangleq \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{k-1} \end{bmatrix} \quad (3.57)$$

Then in Equation (3.56), we note that $\Psi_{k-1} \in \Re^k$, $Q \in \Re^{k \times n}$, and $\mathbf{x}(0) \in \Re^n$. So to solve the equation

$$\Psi_{k-1} = Q\mathbf{x}(0)$$

for the initial condition $\mathbf{x}(0)$, we must have $\Psi_{k-1} \in R(Q)$. Unlike in the previous example, we want a unique solution in this example, because the system can only have started in a single initial state. The condition for such a unique solution is

$$r([Q \ \vdots \ \Psi_{k-1}]) = r(Q) = n \quad (3.58)$$

which, of course, will again require that $k \geq n$. If we take $k = n$, and if $r(Q) = n$, then because $\Psi_{n-1} \in \Re^n$, we will have $r([Q \ \vdots \ \Psi_{k-1}]) = r(Q)$. In that situation, then, the critical question is whether $r(Q) = n$ for Q defined in (3.57). A system that satisfies this criterion is said to be *observable*. See Chapter 8.

3.3.2 Least-Squared Error Solutions and Experimental Data Analysis

A by-product of the solution to the overdetermined problem is the solution of a related problem: finding the vector $\mathbf{x}$ that is *closest* to being a solution to $\mathbf{y} = A\mathbf{x}$ where no exact solution exists, i.e., when $r(A) \neq r(W)$. We may wish to know the vector that minimizes the squared-error criterion (3.48). That vector will be the orthogonal projection of $\mathbf{y}$ onto the subspace of $\Re^m$ spanned by the columns of matrix A , because $\mathbf{y}$ itself does not lie in this subspace. Because the derivation resulting in the pseudoinverse solution of Equation (3.49) does not require that the error $\mathbf{e} = \mathbf{y} - A\mathbf{x}$ be zero, the same formula will serve to find this *least-squared error* solution.

A common situation in which we might seek a least-squared solution is when experimental data is gathered from a physical process. The process might be truly linear, such as the relationship between position and constant velocity, $x = vt + x_0$, or it may be an unknown process that we wish to model as a linear

process, such as when we perform linear regression on a set of data points. In either of these situations, inaccuracies in measurements and unmodeled effects will often prevent the set of linear equations from having a solution, according to the criterion $r(A) = r(W)$. Nevertheless, we want a solution and a measure of the error introduced by that solution. Such a solution is given by Equation (3.49) above.

For example, suppose we have control of two experimental parameters a_1 and a_2 . Suppose further that we suspect that a physically observed quantity y depends linearly on these two parameters as $a_1x_1 + a_2x_2 = y$. If we perform m experimental trials, changing a_1 and a_2 each time and recording the resulting output y , we will gather the following data:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 &= y_1 \\ a_{21}x_1 + a_{22}x_2 &= y_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 &= y_m \end{aligned}$$

which can be written in vector-matrix notation as

$$A\mathbf{x} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ \vdots & \vdots \\ a_{m1} & a_{m2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} = \mathbf{y}$$

If m is large because many experimental trials were performed, then any noise in the measurements will make it extremely unlikely that $r(A) = r(W)$. This would require that a vector of many components be a linear combination of only two vectors, i.e., a member of a two-dimensional space. Even if there is little or no noise in the data, numerical inaccuracies inherent in the computation of matrix rank can lead to the conclusion that there is no exact solution. In such a case, we use the least-squared error solution in (3.49) and the error equation (3.47) to provide an accurate measurement.

Example 3.12: Analyzing Data All at Once

A student in the networks laboratory analyzes an amplifier circuit and determines that, theoretically, the observed output current i_{out} should be proportional to a linear combination of two input voltages, v_x and v_y . She then applies two variable voltage sources as v_x and v_y and records the current i_{out} for each set of inputs. The data she records is shown in the table below:

v_x	-5	-5	-5	-5	-5	-2	-2	-2	...
v_y	-5	-2	0	2	5	2	0	5	...
i_{out}	17.5	-3.5	-17.5	-32	-50	-21	-7	-42	...

v_x	0	0	2	2	2	5	5	5
v_y	2	5	-5	0	5	-5	0	5
i_{out}	-14	-35	42	7	-28	52.5	15	-17.5

Find the unknown gain constants in the following model for the relationships between the variables.

$$i_{out} = g_1 v_x + g_2 v_y \quad (3.59)$$

Solution:

We can model the descriptive equation in vector-matrix form as:

$$i_{out} = \begin{bmatrix} v_x & v_y \end{bmatrix} \begin{bmatrix} g_1 \\ g_2 \end{bmatrix}$$

There are a total of 16 input/output data sets, so we have

$$\mathbf{i}_{16 \times 1} = \mathbf{V}_{16 \times 2} \mathbf{g}_{2 \times 1}$$

Using a computer to perform rank tests, we find that $r(V) = 2$, and $r(\begin{bmatrix} V \\ \mathbf{i} \end{bmatrix}) = 3$, so there is no exact solution. The best solution that can be found is the least-squared error solution provided by

$$\mathbf{g} = (\mathbf{V}^T \mathbf{V})^{-1} \mathbf{V}^T \mathbf{i} = \begin{bmatrix} 3.4 \\ -7.0 \end{bmatrix} \quad (3.60)$$

(to two significant digits).

Analyzing the error in this result, we find that

$$\|\mathbf{e}\| = \|\mathbf{i} - \mathbf{V}\mathbf{g}\| = 3.2$$

To see visually the accuracy of the plane given by Equation (3.59) with the

estimated parameters g_i from Equation (3.60), a plot of the estimated plane (3.59) and the data points given in the problem statement are shown in Figure 3.11. Clearly, the parameters g_i give a reasonable fit to the data.

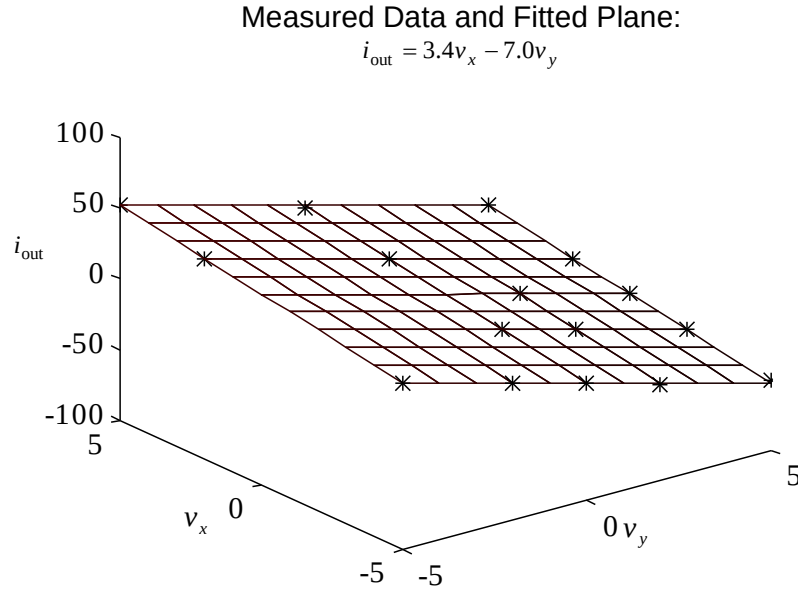


Figure 3.11 Measured data (*) and fitted plane for Example 3.12.

Sequentially Processed Data: Recursive Least-Squares

Note that in the example above, the matrix A will grow larger as m increases, i.e., as more data is taken. Because of this, the experimenter is required to store all the data in a single array and then compute the pseudoinverse on this large array (matrix). If it is then desired that more data points be taken, perhaps because the error e is unacceptably high, the old array must be retrieved, augmented with new rows representing new data, and the pseudoinverse recomputed, disregarding the information provided by the previously performed pseudoinverse. Rather than “waste” this previous calculation, a recursive formulation of the least-squared error solution can be derived.

After some minimal number k of data points has been gathered, then an initial computation will produce a solution $\mathbf{x}_k = (A_k^T A_k)^{-1} A_k^T \mathbf{y}_k$. If, in the example above, the first $k = n = 2$ trials were performed with linearly independent sets of a -parameters, then a simple inverse will provide an initial solution, $\mathbf{x}_k = A_k^{-1} \mathbf{y}_k$. When the $(k+1)^{\text{st}}$ data point is collected, we can augment the A -matrix with a new row, denoted a :

$$A_{k+1} = \begin{bmatrix} A_k \\ \vdots \\ a \end{bmatrix}$$

Then the pseudoinverse solution can be written as

$$\begin{aligned} \mathbf{x}_{k+1} &= \left(A_{k+1}^T A_{k+1} \right)^{-1} A_{k+1}^T \mathbf{y}_{k+1} \\ &= \left(\begin{bmatrix} A_k^T & \vdots & a^T \end{bmatrix} \begin{bmatrix} A_k \\ \vdots \\ a \end{bmatrix} \right)^{-1} \begin{bmatrix} A_k^T & \vdots & a^T \end{bmatrix} \begin{bmatrix} \mathbf{y}_k \\ \vdots \\ y_{k+1} \end{bmatrix} \\ &= \left(A_k^T A_k + a^T a \right)^{-1} \left[A_k^T \mathbf{y}_k + a^T y_{k+1} \right] \end{aligned} \quad (3.61)$$

For help inverting the sum of matrices inside the parentheses above, we use the following matrix identity known as the matrix inversion lemma (see also Appendix A):

LEMMA: For compatibly dimensioned matrices B , C , D , and E , the following identity holds:

$$\left(B^{-1} + CDE \right)^{-1} = B - BC \left(EBC + D^{-1} \right)^{-1} EB \quad (3.62)$$

For our situation, we can make the identifications $B^{-1} \Leftrightarrow A_k^T A_k$, $C \Leftrightarrow a^T$, $D \Leftrightarrow I$, and $E \Leftrightarrow a$. Therefore, by direct substitution,

$$\begin{aligned} \mathbf{x}_{k+1} &= \left\{ \left(A_k^T A_k \right)^{-1} - \left(A_k^T A_k \right)^{-1} a^T \left[a \left(A_k^T A_k \right)^{-1} a^T + I \right]^{-1} a \left(A_k^T A_k \right)^{-1} \right\} \\ &\quad \cdot \left[A_k^T \mathbf{y}_k + a^T y_{k+1} \right] \end{aligned} \quad (3.63)$$

By comparison with (3.61), we see that the quantity in braces in (3.63) is equal to $\left(A_{k+1}^T A_{k+1} \right)^{-1}$, so at the $(k+2)^{\text{nd}}$ sample, it will be used as $\left(A_k^T A_k \right)^{-1}$ is in (3.61). The quantity in braces has already been computed in the previous iteration, so the only inverse that is actually computed in (3.63) is the term in square brackets in the center. As long as only a single additional row of data is collected in each sample, this bracketed term will be a scalar, thus requiring no additional matrix inversions at all. Similarly, the bracketed term at the end, $[A_k^T \mathbf{y}_k + a^T y_{k+1}]$, will become $A_{k+1}^T \mathbf{y}_{k+1}$ for the next computation. No matrices will grow in size.

Equation (3.63) is thus the *recursive least-squared error* solution and is a convenient way to incorporate new information into previously computed results.

Example 3.13: Recursive Least-Squares

Suppose that we want to find the constants g_1 and g_2 in the equation

$$y = g_1 x_1 + g_2 x_2$$

given the set of measured data

input		output
x_1	x_2	y
x_{11}	x_{21}	y_1
x_{12}	x_{22}	y_2
x_{13}	x_{23}	y_3
x_{14}	x_{24}	y_4
$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$

We will demonstrate the algorithm by which the recursive least-squares formulation (3.63) generates estimates of these parameters. Because we have two measured variables, x_1 and x_2 , and two unknown parameters, we can form an initial estimate after only two measurements, i.e., by using the first two lines of the table above. If only the first two measurements are used, then

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_{11} & x_{21} \\ x_{12} & x_{22} \end{bmatrix} \begin{bmatrix} g_1 \\ g_2 \end{bmatrix}$$

has the solution

$$\mathbf{g}_0 = \begin{bmatrix} x_{11} & x_{21} \\ x_{12} & x_{22} \end{bmatrix}^{-1} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \triangleq \mathbf{A}_1 \mathbf{y}_1$$

Now define the starting points for the recursive loop that results from (3.63). Assign

$$\mathbf{A} := (\mathbf{A}_1^T \mathbf{A}_1)^{-1} \quad \text{and} \quad \mathbf{B} := \mathbf{A}_1^T \mathbf{y}_1$$

Because the recursion starts with the *third* row of measured data from the table, set $i = 3$ and then execute the pseudocode algorithm shown here:

```

LOOP {
  set  $a := i^{\text{th}}$  row of input data,  $[x_{1i} \ x_{2i}]$  ;get new independent variables (input)
  set  $y := i^{\text{th}}$  row of output data,  $y_i$  ;get new output data (measurement)
  set  $A := A - Aa^T(aAa^T + 1)^{-1}aA$  ;compute term in braces from (3.63)
  set  $B := B + a^Ty$  ;compute bracketed term from (3.63)
  compute  $x_i = AB$  ;compute an updated estimate
   $i := i + 1$  ;increment measurement count
} ;continue until all the data is used

```

Note that in this algorithm, the terms referred to as A and B , which are, respectively, the term in braces and the term in brackets in (3.63), are recursively assigned. Note also that the inversion performed in the fourth line is scalar inversion.

3.4 Summary

Again, as in Chapter 2, this chapter has presented some of the mathematical basics necessary for understanding state spaces. Of primary importance here was the concept of a linear operator. Rather than thinking in terms of matrix-vector multiplication as used in matrix theory, we present the linear operator approach. This gives a much more geometrical understanding of some of the basic arithmetic computations we have been performing all along. Without this approach, certain linear system theory and control system concepts cannot be fully understood. Although we are perhaps premature in introducing them, the controllability and observability examples were presented in this chapter for just that purpose: to show how the operator interpretation of linear equations leads to a general test for these properties of state space systems. Solutions of linear equations through numerical methods or gaussian elimination cannot lend this insight.

Other important concepts introduced in this chapter were:

- All linear operators can be written as matrices, and all matrices can be thought of as operators. So although we stress the geometric understanding of linear operators, it should be clear that our old computational tools and habits are still useful.
- Linear operators themselves are written in specific bases, either explicitly defined or understood from their context. These bases may be changed with matrix multiplication operations, i.e., Equation (3.24).
- Operators themselves (equivalently, their matrix representations) form vector spaces and have norms of their own. Adjoint operators are the

operator-theoretical manifestations of symmetric and hermitian matrices.

- The process of solving linear algebraic equations can be seen as the search for a preimage of a given vector in the range space of an operator. If the given vector is not in the range space of the operator, least-squared error solutions can be obtained with the projection operator, resulting in a pseudoinverse. In case there are many solutions, a similar pseudoinverse gives the smallest solution.
- Recursive solutions to experimental data analysis problems were presented for two reasons: they will be familiar to readers who continue to Chapter 11, which includes recursive estimators. Also, they are efficient tools for processing real-time data.

Up to this point, the mathematical preliminaries we have discussed have intentionally been kept rather general. Beginning in the next chapter, we will restrict our discussion of vector spaces and linear operators to state spaces and operators that operate on, and into, the same state space. Such operators will then be used to help solve and understand the state space equations developed in Chapter 1.

3.5 Problems

- 3.1 Determine which of the following operations on vectors are linear operators:

a) $A\mathbf{x} = \mathbf{x} + \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$, where $\mathbf{x} \in \mathfrak{R}^3$

b) $A\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, where $\mathbf{x} \in \mathfrak{R}^3$

c) $A\mathbf{x} = \int_{-\infty}^{\infty} f(\sigma)\mathbf{x}(t - \sigma)d\sigma$, where $f(\cdot)$ is a continuous function, and $\mathbf{x}$ is in the vector space of continuous functions.

d) $A\mathbf{x} = \begin{bmatrix} x_1^2 \\ x_2^2 \\ x_3^2 \end{bmatrix}$, where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

$$\text{e) } A\mathbf{x} = \begin{bmatrix} \sin x_1 \\ \cos x_2 \\ 0 \end{bmatrix}, \text{ where } \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\text{f) } A\mathbf{x} = \begin{bmatrix} x_1 + 2x_2 + 3x_3 \\ x_1 + x_2 + x_3 \\ x_1 \end{bmatrix}, \text{ where } \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$3.2 \quad \text{Show that } \frac{d(A^{-1}(t))}{dt} = -A^{-1}(t) \frac{dA(t)}{dt} A^{-1}(t).$$

3.3 It is known that if $n \times n$ matrices A^{-1} and B^{-1} exist, then $(AB)^{-1} = B^{-1}A^{-1}$. Does $(A+B)^{-1}$ necessarily exist?

3.4 Let A be a linear operator on the space $\mathfrak{R}^3$ that transforms the vectors

$$\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\} = \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \right\}$$

into vectors

$$\{\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3\} = \left\{ \begin{bmatrix} -2 \\ 4 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ -1 \end{bmatrix} \right\}$$

respectively, i.e., $A\mathbf{x}_i = \mathbf{y}_i$ for $i = 1, 2, 3$.

- a) Find the matrix representation of A in the standard basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$.
- b) Find the matrix representation of A in the basis consisting of the vectors $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$.

3.5 Consider two bases for space $\mathfrak{R}^3$:

$$\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\} = \left\{ \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ -2 \\ -2 \end{bmatrix}, \begin{bmatrix} -4 \\ 2 \\ 1 \end{bmatrix} \right\} \quad \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\} = \left\{ \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -4 \\ -3 \\ -2 \end{bmatrix}, \begin{bmatrix} 5 \\ -2 \\ 0 \end{bmatrix} \right\}$$

If a linear operator is given in the $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ basis by

$$A = \begin{bmatrix} 8 & -2 & -1 \\ 4 & -2 & -3 \\ 2 & -3 & -3 \end{bmatrix}$$

find the representation for this operator in the basis $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$. If we are given a vector $\mathbf{x} = [2 \ -1 \ -4]^T$ in the basis $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$, determine the representation for $\mathbf{x}$ in the basis $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$.

3.6 Consider the vectors in $\mathfrak{R}^3$:

$$\mathbf{x}_1 = \begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix} \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \quad \mathbf{x}_3 = \begin{bmatrix} 8 \\ 5 \\ -7 \end{bmatrix}$$

and the vectors in $\mathfrak{R}^4$:

$$\mathbf{y}_1 = \begin{bmatrix} 1 \\ 3 \\ -1 \\ 0 \end{bmatrix} \quad \mathbf{y}_2 = \begin{bmatrix} 2 \\ 2 \\ 1 \\ 2 \end{bmatrix} \quad \mathbf{y}_3 = \begin{bmatrix} 5 \\ -6 \\ 3 \\ 10 \end{bmatrix}$$

where $T\mathbf{x}_i = \mathbf{y}_i$, for $i = 1, 2, 3$.

- Write the matrix of the operator $T: \mathfrak{R}^3 \rightarrow \mathfrak{R}^4$ in the basis $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ for $\mathfrak{R}^3$ and the standard basis for $\mathfrak{R}^4$.
- Write the matrix of the operator T in the standard basis in $\mathfrak{R}^3$ and the standard basis in $\mathfrak{R}^4$.
- Find the rank and degeneracy of T .
- Find a basis for the null space of T .
- Find a basis for the range space of T .

3.7 The matrix of a linear operator $T: \mathfrak{R}^3 \rightarrow \mathfrak{R}^3$ given in the standard basis

$$\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \text{ (for both the domain and the range) is}$$

$$T = \begin{bmatrix} 12 & -2 & 1 \\ -3 & -5 & 2 \\ 4 & -6 & -4 \end{bmatrix}$$

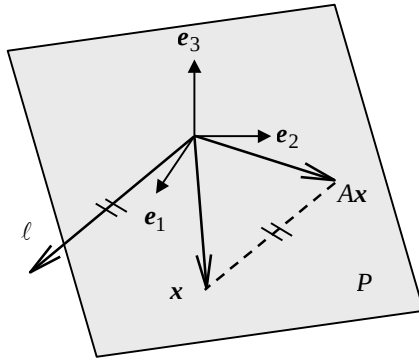
Find the matrix for this operator if both bases are

$$\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\} = \left\{ \begin{bmatrix} -3 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ -2 \\ 2 \end{bmatrix} \right\}$$

3.8 Consider linear operator $A: \mathbf{X} \rightarrow \mathbf{Y}$. Use the definition of a subspace to show that null set of A , i.e., the set of all vectors $\{\mathbf{x} \mid A\mathbf{x} = 0\}$ is indeed a subspace of $\mathbf{X}$.

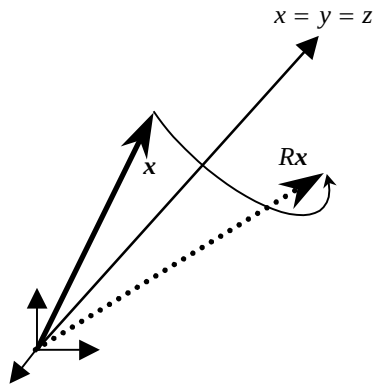
3.9 Let P be the plane in $\mathfrak{R}^3$ defined by $2x + 2y + 1z = 0$, and let ℓ be the vector $\ell = [8 \ -2 \ 2]^T$. Denote by $A: \mathfrak{R}^3 \rightarrow \mathfrak{R}^3$ the projection operator that projects vectors in $\mathfrak{R}^3$ onto the plane P , not orthogonally, but along the vector ℓ (that is, the projections can be pictured as “shadows” of the original vectors onto P , where the light source is at an infinite distance in the direction of ℓ). See Figure P3.9.

- Find the matrix of operator A in the basis $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3\}$, where $\mathbf{f}_1$ and $\mathbf{f}_2$ are any noncollinear vectors in the plane P , and $\mathbf{f}_3 = \ell$.
- Find the matrix of operator A in the standard basis of $\mathfrak{R}^3$.



P3.9

- 3.10 Let $R: \mathfrak{R}^3 \rightarrow \mathfrak{R}^3$ denote the three-dimensional rotation operator that rotates vectors by an angle $\pi/3$ radians about the axis $x = y = z$ in $\mathfrak{R}^3$ (counterclockwise as viewed by an observer at $[1 \ 1 \ 1]^T$ looking back at the origin). See Figure P3.10.



P3.10

- Show that R is a linear operator.
- Find the matrix representation for R in the basis $\{f_1, f_2, f_3\}$, which are mutually orthogonal unit vectors, and f_3 is in the direction of the axis of rotation.

- c) Find the matrix representation for R in the standard basis.
- 3.11 Consider the space P of polynomials of order less than four, over the field of reals. Let $D: P \rightarrow P$ be a linear operator that takes a vector $\mathbf{p}(s)$ and returns $\mathbf{p}''(s) + 3\mathbf{p}'(s) + 2\mathbf{p}(s)$.
- a) Find the matrix representation of D in the basis $\{1, s, s^2, s^3\}$.
- b) Find the matrix representation of D in the basis $\{s+1, 2s+3, s^2+5s+4, s^3+s\}$.
- 3.12 The orthogonal complement $A^\perp$ of a linear space A is defined as being the set $\{\mathbf{x} \mid \langle \mathbf{x}, \mathbf{y}_i \rangle = 0 \text{ for all } \mathbf{y}_i \in A\}$, i.e., every vector in $A^\perp$ is orthogonal to every vector in A . Show that $N^\perp(A) = R(A^*)$ and $R^\perp(A) = N(A^*)$, where A^* is the adjoint operator of A .
- 3.13 Let $A = \begin{bmatrix} 2 & -2 \\ 1 & 4 \end{bmatrix}$. Find all 2×2 matrices X such that $AX = 0$.
- 3.14 For the operators represented by the matrices given in Problem 2.10, find bases for the null space and the range space.
- 3.15 For the pair of matrices

$$A = \begin{bmatrix} 2 & 3 & 1 & 4 & -9 \\ 1 & 1 & 1 & 1 & -3 \\ 1 & 1 & 1 & 2 & -5 \\ 2 & 2 & 2 & 3 & -8 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 17 \\ 6 \\ 8 \\ 14 \end{bmatrix}$$

determine all possible solutions to the system $A\mathbf{x} = \mathbf{y}$.

- 3.16 Determine the values of α such that the system of linear equations

$$\begin{aligned} 3x_1 + 2x_2 + x_3 + 9x_4 &= 4 \\ -4x_1 + 2x_2 - 6x_3 - 12x_4 &= -3 + 7\alpha \\ x_1 + 4x_2 - 3x_3 + 3x_4 &= -2 \end{aligned}$$

has exact solutions, and find the general solution to the system.

- 3.17 Find the best solution, in the least-squared error sense, to the system of equations:

$$\begin{aligned} -2 &= x_1 - 2x_2 \\ 5 &= x_1 - 2x_2 \\ 1 &= -2x_1 + x_2 \\ -3 &= x_1 - 3x_2 \end{aligned}$$

- 3.18 Determine whether the three lines intersect at a common point

$$\begin{aligned} x - 2y &= -5 \\ 4x - 1y &= 8 \\ 2x &= 6 \end{aligned}$$

- 3.19 Find the particular and homogeneous solutions to the system of equations:

$$\begin{bmatrix} 2 & 2 & -3 \\ 4 & 8 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2 \\ -12 \end{bmatrix}$$

- 3.20 Determine whether the following system has an exact solution, and if so, find it.

$$\begin{aligned} x_1 + 5x_2 + 7x_3 + 4x_4 &= 7 \\ 2x_1 + 4x_2 + 8x_3 + 5x_4 &= 2 \\ 3x_1 + 3x_2 + 9x_3 + 6x_4 &= -3 \end{aligned}$$

3.21 Use the recursive least-squares method (see page 133) to solve Example 3.12.

3.22 If the *unweighted* pseudoinverse solution

$$\mathbf{x} = A^+ \mathbf{y} \triangleq A^T (AA^T)^{-1} \mathbf{y}$$

minimizes the value of $\frac{1}{2} \mathbf{x}^T \mathbf{x}$ subject to the constraint $A\mathbf{x} = \mathbf{y}$, what does the *weighted* pseudoinverse solution

$$\mathbf{x} = A_B^+ \mathbf{y} \triangleq BA^T (ABA^T)^{-1} \mathbf{y}$$

minimize under the same constraint?

3.6 References and Further Reading

Our perspective on linear operators as given here is a reflection of references [12], [13], and [14]. These sources provide a background on operators in a general setting, rather than simply as matrices. While [12] and [13] are somewhat more mathematical (and [12] includes some material on topology and metric spaces), [14] offers a more applied approach. For computational references, see [6] and [7].

Our approach to projections and their applications is similar to [3] and is motivated by their applications in approximation theory and optimization theory. See [4] for further details on the use of projections in optimization.

The idea of an adjoint operator can be further pursued in [2] and [8]. This will lead to the concept of an *adjoint system*, which has implications in the duality of control systems concepts such as controllability and observability (Chapter 8), and regulators and observers (Chapter 10).

For further information in system identification with input/output data, [10] is a standard reference, but it does not emphasize the recursive least-squares approach that we introduce here. For such a treatment, see [9]. For further details on the various types and applications of generalized inverse matrices (including pseudoinverses), see [1].

The robotics applications can be pursued in [5] and [11]; the latter emphasizes the redundancy problem in robot kinematics, in which an underdetermined system must be solved.

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